



Socially-Aware Joint Computation Offloading and Bandwidth Allocation for Multi-access Edge Computing

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Socially-Aware Joint Computation Offloading and Bandwidth Allocation for Multi-access Edge Computing

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Abstract—Multi-access edge computing (MEC) provides powerful computational support for the Internet of Things (IoT), accelerating the processing of computational tasks while avoiding the latency associated with connecting to a central server. However, due to the limited computational resources of edge nodes and end devices, it is crucial to design effective resource allocation strategies. Moreover, in the process of edge computing services, individual user behavior is typically considered, without taking into account the interdependencies between them. In this paper, we propose a Bayesian Stackelberg game with incomplete information to analyze the interactions between edge service providers and end devices, while also considering the social attributes of end users. The edge service provider acts as the leader, providing edge computing services, while the end users act as followers, requesting computing services. Both parties aim to maximize their benefits. In the task offloading process, the transmission energy consumption of end users is reduced by establishing an optimal relationship between bandwidth allocation and offloading task size. Additionally, the Stackelberg equilibrium (SE) is derived to verify the existence and uniqueness of the SE, and a gradient-based solution algorithm is proposed. The simulation results validate the effectiveness of the proposed Bayesian Stackelberg game.

Index Terms—Computing offloading, multi-access edge computing, Stackelberg game, incomplete information, resource allocation.

I. INTRODUCTION

With the rapid development of the Internet technologies and the widespread adoption of smart devices [1], the Internet of Things (IoT) has been extensively applied, leading to a rapid increase in various service applications within the network. Consequently, a number of computation-intensive and latency-sensitive applications have emerged [2], such as real-time video processing, autonomous driving, and virtual reality. However, due to the typically limited computational capabilities of IoT devices [3], they are often unable to meet the service demands of these applications. Although cloud computing can process and store large volumes of data, its centralized architecture and the physical distance from end-user devices may hinder its ability to satisfy the required

quality of service (QoS). To address this issue, multi-access edge computing (MEC) has emerged as a promising solution. Compared to cloud computing, MEC servers are deployed closer to end users, allowing mobile devices to offload data without the need to transmit it to distant cloud servers, thereby significantly reducing communication latency [4]. By providing cloud-like computing and storage capabilities at the edge of the network, MEC not only conserves energy for end-user devices but also greatly reduces service latency [5].

Since MEC servers do not possess the sufficient computing and storage capacities of traditional cloud computing systems, it is therefore crucial to design effective computation offloading mechanisms to optimize the quality of network services. Edge networks may suffer service pressure due to large uploads of computationally intensive tasks, resulting in tasks not being completed within the required time. These problems are not uncommon and may occur more frequently due to network congestion [7]. Consequently, designing an effective computation offloading mechanism is essential for optimizing the quality of network services throughout the entire service lifecycle. In recent years, resource allocation and task offloading in MEC systems have been extensively studied [6] [8] [9] [10] [11]. By optimizing network resource allocation and task offloading strategies, the overall performance and quality of service of edge computing systems can be significantly improved, which helps alleviate the service pressure on the central cloud and satisfy low-latency application requirements.

To further improve network service quality, the social perception factors between user devices can be considered [12] [13]. Research based on social networks shows that there is a close relationship between the social relationships of users and their behavior [14] [15]. Due to the presence of social relationships, end users are concerned not only with their utility but also with the utility of their neighbors within their social circles. Due to the existence of social relationships, end users are not only concerned about the utility they obtain for themselves, but also pay attention to the utility of their neighbors in their social circle. Inspired by this, we can leverage the social connections between users to formulate mutually beneficial computing offloading decisions, thereby enhancing the overall service performance of edge computing network systems. Therefore, a key challenge in MEC systems is how to achieve reasonable and efficient computing offloading and resource allocation while improving the profits of edge service providers and the offloading utility of end users based on social relationships between users.

Meanwhile, in order to meet the growing service demands of various applications, edge service providers enhance their

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computing resources by deploying additional edge servers to ensure the efficient and secure offloading of tasks. However, the expansion of server deployment inevitably leads to increased costs. To reduce service costs, edge service providers may increase service prices. However, a higher service price may lead users to lower their service demand, thereby reducing overall service revenue [16]. Therefore, it is also crucial to design a resource pricing strategy to safeguard the economic interests of both sides in edge computing services.

In this paper, we propose a Stackelberg game model to analyze the computation offloading interaction process between edge service providers and end users in MEC. Specifically, in the game model, edge service providers are considered leaders, deciding the pricing of computing resources. End users are considered followers, deciding the task offloading size. For edge service providers, social interaction information from other IoT users can offer constructive evaluations for potential service provision. At the same time, such information can assist end users in making informed decisions. Considering the incompleteness of social interaction information in real-world scenarios, we construct a Stackelberg game model under incomplete information. We further verify the existence and uniqueness of the Nash equilibrium. Finally, a dynamic iterative algorithm is used to obtain the Nash equilibrium solution of the game, ensuring that edge service providers and end users obtain the optimal service utility. The main contributions of this paper are as follows.

- We propose a single-leader, multi-follower Stackelberg game to analyze the interaction between edge service providers and end users, where end users act as followers and edge service providers as leaders. Specifically, in this game model, social information is incorporated into the utility function of end users, and the energy consumption benefits of task offloading are taken into account. By optimizing the computational offloading strategy and uplink bandwidth allocation strategy for end users, the resource pricing of the edge service provider, the model aims to jointly maximize the service revenue of the provider and the utility of end users.
- Considering the incompleteness of social interaction information among end users, we designed a computational offloading and resource allocation algorithm based on Bayesian Stackelberg games. Meanwhile, to reduce the transmission energy consumption of end users, we consider optimizing the uplink bandwidth allocation strategy for end users. This is achieved by transforming the original problem into an independent subproblem that can be solved independently, and then solving it using the Karush–Kuhn–Tucker (KKT) conditions to obtain the optimal bandwidth allocation strategy. Additionally, we analyze the optimal strategies of all game participants and prove the existence and uniqueness of the Stackelberg game equilibrium, as well as the convergence and complexity of the algorithm.
- Numerous simulation experiments were conducted to validate the effectiveness of the proposed socially-aware and privacy-preserving task offloading method. The sim-

ulation results demonstrate that the proposed approach is capable of achieving Stackelberg equilibrium strategies under varying levels of social relationships. Moreover, it significantly enhances the utilities of both end users and edge service providers. Additionally, the results further confirm that the proposed method exhibits fast convergence and high scalability.

The rest of this paper is organized as follows: related work is introduced in Section II; Section III introduces the system model and formulation of the optimization problem; Section IV analyzes the proposed game model and provides a solution algorithm; Section V presents simulation results and analysis; and finally Section VI concludes the paper.

II. RELATED WORK

Currently, the joint optimization of network resource allocation and task offloading in MEC has attracted extensive research attention. To ensure task latency and computing service requirements for users, Liu et al. [17] jointly optimize service caching, task offloading, computational resources, and bandwidth allocation to minimize the system's service latency and caching cost. To enable MEC servers to provide more real-time computing services to terminals, Liu et al. [18] achieve the minimization of the total request completion latency in MEC networks through joint optimization of request assignment and computational resource allocation. With the integration of a large number of heterogeneous devices into MEC, network resource management becomes increasingly complex. Gao et al. [19] proposed a computation offloading scheme based on attention-weighted recurrent multi-agent critics to meet the computational service demands of large-scale heterogeneous devices. Wang et al. [20] considered a multi-user and multi-server scenario, achieving minimization of energy consumption, response time, and cost through joint optimization of task offloading, power allocation, and resource allocation. Considering the limited computational resources of a single unmanned aerial vehicle (UAV) in a UAV-assisted MEC network, Tun et al. [21] proposed a multi-UAV-assisted collaborative MEC system that minimizes system service latency by optimizing task offloading, communication resources, and computational resources.

Game theory, as a framework for analyzing decision-making behaviors characterized by interdependence and interaction among decision-makers, has been widely applied to optimize network resource allocation in MEC. Wu et al. [22] considered inter-cell interference in edge-cloud computing and proposed a novel multi-cell network computation offloading scheme based on potential games for edge-cloud collaboration, aiming to minimize the total cost in terms of latency and energy consumption. Pham et al. [23] addressed the computation offloading problem in multi-carrier non-orthogonal multiple access (NOMA) enhanced MEC systems and proposed a coalition formation game-based task offloading approach from the perspective of cooperative game theory to optimize network service performance. Gao et al. [24] aimed to improve the task processing efficiency of edge servers by introducing a game theory-based approach and designing a billing mechanism

with instantaneous load, thereby optimizing the order of task offloading. To alleviate the service pressure on edge servers, Lin et al. [25] considered the profit of service providers and proposed a multi-stage coalition game-based task offloading framework to jointly maximize service provider profits and minimize end-to-end delay for service users. To achieve load balancing in mobile edge computing systems, Wu et al. [26] developed a load balancing algorithm based on mean field games, which reduces task processing latency and thereby facilitates task scheduling. Zhang et al. [27] studied a three-tier multi-user multi-task offloading problem with selfish individuals and designed a distributed resource allocation algorithm based on a Stackelberg game to enhance the benefits of both users and edge nodes.

Although the studies above have all examined resource allocation issues in MEC networks, most research has focused solely on the service objectives of users or edge servers during the computing process. Different from the work above, this paper integrates market-based pricing models and game theory-based pricing models. We use the Stackelberg game model to describe the relationship between the edge service provider and end users, and provide services through resource pricing strategies for different user demands. Meanwhile, we consider the social attributes between users when formulating the task offloading strategy for end users. In addition, we optimize bandwidth allocation strategies to reduce task offloading energy consumption for end users.

III. SYSTEM MODEL AND PROBLEM FORMULATION

In this section, we first introduce the network system model. Then, we introduce the utility function model. Finally, we formulate a Stackelberg game to analyze the service relationship between the edge service provider and end devices.

A. System Model

In this paper, we consider computational offloading in a social IoT scenario, as shown in Fig. 1. We assume that there is an edge service provider and N end users in the system. Consider that each end device has some computational tasks to be processed, which it can either process locally or rent free edge resources within its range to help it with its computational tasks. Edge service providers can increase their utility by handling offloaded computational tasks, and end users reduce their costs and computational stress by offloading computation.

Each end user can offload tasks can be offloaded on the edge service provider via a wireless link. It is worth noting that this paper only considered one edge service provider. The end user i to edge service provider transmission data rate r_i can be calculated as [28]

$$r_i = b_i B \log \left(1 + \frac{p_i g_i}{\sigma^2} \right), \quad (1)$$

where B denotes the channel bandwidth, b_i denotes bandwidth allocation strategy, p_i denotes the transmission power of the end device i , g_i denotes the channel gain, and σ denotes the channel noise power.

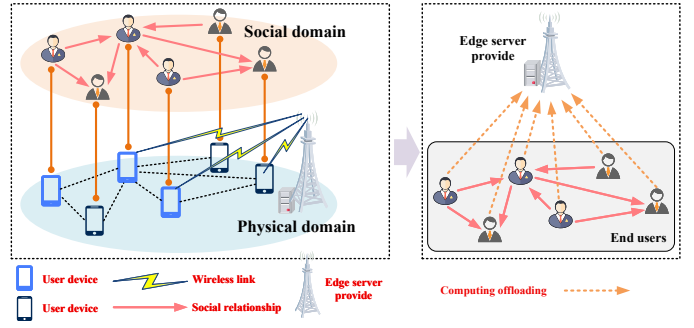


Fig. 1: system model.

Assume that end user i has a computational task $T_i = \{D_i, C_i, t_i^{max}\}$, where D_i denotes the data size of the computational task, C_i denotes the CPU cycles required per unit task computation, and t_i^{max} denotes the maximum tolerant delay for task T_i . Depending on the task T_i , the task can be divided into various types of tasks. We assume that the user's computation task is divisible [29], and the user can choose to execute part of the task locally and offload the remaining part of the task to the edge node to complete the computation.

Let $X = \{x_1, \dots, x_n\}$, $n \in N$ denote the offloading strategies of all end users. The size of the task that needs to be processed locally by end user i is $D_i - x_i$. The local computing latency can be expressed as

$$t_i^{loc} = \frac{(D_i - x_i) C_i}{F_i}, \quad (2)$$

where F_i denotes the local computing capacity of end user i . For the task of offloading part of the computation for end user i , the transmission delay can be represented as

$$t_i^{tran} = \frac{x_i}{b_i B \log \left(1 + \frac{p_i g_i}{\sigma^2} \right)}. \quad (3)$$

The transmission energy consumption of end user i can be represented as $E_i^e = p_i t_i^{tran}$.

Let f_i^e denote the computing resources allocated by the edge service provider to user i . To facilitate the analysis of computing task latency at the edge, we assume that computing resources at the edge are uniformly distributed, i.e., computing resources are allocated using a uniform allocation strategy [30]. Therefore, the computing resources available to end users i can be expressed as $f_i^e = f^{edge}/N$, where f^{edge} represents the total computing resources of the edge service provider. Although the uniform distribution strategy is relatively simple, it can meet the needs of most edge computing scenarios [31] [32]. The computing time of the task in edge service providers can be expressed as

$$t_i^{comp} = \frac{x_i C_i}{f_i^e}. \quad (4)$$

The total latency for task offloading for end user i can be expressed as

$$t_i^e = t_i^{tran} + t_i^{comp}. \quad (5)$$

B. Utility Function Model

In this paper, users compute tasks through the resources of the edge service provider, and the edge service provider must ensure that the available resources can satisfy the user demand. To adjust the demand and supply of edge resources, the edge service provider can adjust the use of resources using resource pricing. Therefore, the Stackelberg game is used to describe the interaction between edge providers and users. We model the interaction between edge service providers and socially aware participants (i.e., end users) as a Stackelberg game, where each end user's behavior is to request appropriate resources, and the edge service provider's behavior is to select appropriate resource pricing to provide computing services to end users. It is worth noting that in the Stackelberg game, the edge service provider is the leader and the end user is the follower.

We consider a single-leader multi-follower Stackelberg game. During the game, the leader (i.e., edge service provider) first sets the price ρ for the computational resource. Subsequently, the end user determines the optimal amount of computing resources required, i.e., offloading policy X , based on the resource pricing of the edge service provider.

1) *The utility function of the end user:* In this paper, we consider the impact of social information in social IoT. The IoT device considers the impact of information from other IoT devices on the task offloading benefits, in addition to its task offloading benefits. The user utility includes task offloading benefits, external benefits to the network, the cost of task offloading, and the reward generated by task offloading. The utility function of the end user can be expressed as,

$$U_i = f_i(x_i) + \Psi(x_i, x_{-i}) + R(x_i) - E(x_i, p_i), \quad (6)$$

where $f_i(x_i, s_i)$ denotes the offloading revenue for the user, $\Psi(x_i, x_{-i})$ denotes the external benefits from the network, x_{-i} denotes all other users except user i , $\Theta(x_i)$ denotes the compensation for privacy loss provided by edge service provider, $R(x_i)$ denotes the reward generated by the task offloading, and $E(x_i)$ denotes the task offloading overhead of the end user, which mainly consists of the communication overhead of the task transmission and the cost of the end user in purchasing the edge resources.

During the resource request process, users tend to buy more resources to increase their revenue to reduce the consumption of their resources. However, with the increase of resources requested by the user, the computational offloading revenue of the task decreases gradually, i.e., the marginal effect on the end user is diminishing. Similar to [35], task offloading revenue can be expressed as

$$f(x) = ax - bx^2, \quad (7)$$

where $a > 0$ and $b > 0$ denote the satisfaction coefficient.

The end user can derive additional benefits from the information contributed or shared by other user devices [36]. Considering the structural properties of the Social IoT, we define an adjacency matrix $G = [v_{ij}]$, $i, j \in N$ for analyzing the influence between users, where v_{ij} denotes the strength of the influence of user j on user i . For the convenience of

analysis, we assume that the adjacency matrix is fixed in this paper. In addition, we assume that the interactions between devices in the social IoT are the same, i.e., $v_{ij} = v_{ji}$. Higher v values indicate a closer relationship between two devices in the Social IoT. Similar to [36], the external benefits of the end user is denoted as

$$\Psi(x_i, x_{-i}) = \sum_{j=1, j \neq i}^N v_{ij} x_i x_j, \quad (8)$$

which denotes the social interaction information that the user obtains from other user devices [37].

In addition, the user purchases computational resources from the edge service provider for the processing of tasks. Therefore, the end user needs to offload the tasks to the edge service provider. The cost of user task offloading includes computing resource purchase cost and task communication cost. The task offloading cost of the end user i is denoted as

$$E(x_i, p_i) = \delta_2 \rho x_i + \delta_3 E_i^c(x_i, p_i), \quad (9)$$

where ρ denotes the resource pricing of the edge service provider, δ_2 and δ_3 denotes the weight coefficients. At the same time, since the user purchases the computing resources of the edge service provider, some tasks will be offloaded to the edge service provider for computation, thus the user reduces the computational spending of the task, i.e., the computational energy consumption of the task is reduced. Therefore, the return generated by user task offloading is expressed as

$$R(x_i) = \delta_1 \kappa F_i^2 C_i x_i, \quad (10)$$

where κ denotes the effective switching capacitance of the edge server [40], δ_1 denotes the weight coefficient.

Then, the utility function of end user i can be expressed as,

$$U_i = ax_i - bx_i^2 + \sum_{j=1, j \neq i}^N v_{ij} x_i x_j + \delta_1 \kappa F_i^2 C_i x_i - \delta_2 \rho x_i - \delta_3 \frac{p_i x_i}{r_i}. \quad (11)$$

2) *The utility function of the edge service provider:* For the edge service provider, the utility function is equal to the profit from the sale of resources minus the computational energy consumption. The utility function of the edge service provider can be expressed as

$$U_e = \sum_{i \in N} (\rho x_i - \varepsilon x_i), \quad (12)$$

where ε denotes the coefficient factor for computing energy consumption. It is worth noting that we consider an edge service provider to adopt differentiated pricing for different users.

C. Problem Formulation

Assuming that end users and edge service providers are both individually rational, both parties aim to maximize their benefits when making decisions. End users will adjust their task offloading size based on resource prices to optimize their utility, and adjust their transmission power to reduce energy

consumption. Edge service providers will consider the demand for computing resources from nearby users to set prices in order to maximize their utility. The utility of the edge service provider depends on the task offloading strategy of end users. In contrast, the resource pricing strategy of the edge service provider influences the offloading strategy of end users. To maximize the utility of both parties, it is necessary to combine the optimal strategy of the edge service provider with the optimal response of end users.

The objective of the end user is to determine the optimal resource request to maximize its utility. Therefore, the optimization problem for the end user can be expressed as,

$$\begin{aligned}
 P1 : & \max_{x_i, p_i} U_i(x_i) \\
 s.t. \quad & C1 : 0 \leq x_i \leq D_i, \\
 & C2 : 0 \leq b_i \leq 1, \\
 & C3 : \sum_{i \in N} b_i \leq 1,
 \end{aligned} \tag{13}$$

where constraint C1 represents the range of offloading task sizes, constraint C2 denotes that the bandwidth resource allocation strategy cannot exceed the channel bandwidth capacity, and constraint C3 denotes that the bandwidth allocation strategy for all users cannot exceed the channel bandwidth capacity. Respectively.

We can describe the utility maximization problem of an edge service provider as,

$$\begin{aligned}
 P2 : & \max_{\rho_i} U_e(\rho_i) \\
 s.t. \quad & C1 : 0 \leq \rho_i \leq \rho^{max},
 \end{aligned} \tag{14}$$

where constraint C1 denotes resource pricing constraints, ρ^{max} denotes the maximum resource pricing of the edge service provider, Constraint C2 denotes the investment ratio constraint for a single user, and constraint C3 denotes the investment ratio constraint for all users.

IV. GAME EQUILIBRIUM ANALYSIS

In practice, the selfish behavior of IoT devices can lead to the inability to obtain individual information from each other due to the lack of global supervision and monitoring. In this case, we utilize information about the distribution of IoT devices, which can be obtained based on historical information. Therefore, incomplete information is more applicable to the scenario considered in this paper, and we formulate the problem as a Bayesian Stackelberg game [41]. Backward induction methods are used to analyze Bayesian Stackelberg games under incomplete information. In this section, we analyze the follower game and the leader game in turn. Before analyzing the equilibrium process of the game, we define the Nash equilibrium solution of the game.

Definition 1: Assume that (x^*, ρ^*) is a Nash equilibrium solution to the game, and for any solution (x, ρ) satisfies $U_i(x_i^*, x_{-i}^*, \rho^*) \geq U_i(x_i, x_{-i}, \rho)$, $i \in N$, and $U_e(x^*, \rho^*) \geq U_e(x^*, \rho)$ where x_{-i} denotes the optimal policy for all other users except user i .

A. Problem Transformation

To simplify the social interaction information, we denote the social factors as $\xi = v_{ij} > 0$, and this component represents the desired social interaction information. Therefore, the expected utility of end user i is denoted as,

$$\begin{aligned}
 U'_i &= E(U_i(x_i, x_{-i})) \\
 &= ax_i - bx_i^2 + \xi x_i E \left[\sum_{j \in N, j \neq i} x_j \right] + \delta_1 \kappa F_i^2 C_i x_i - \\
 &\quad \delta_2 \rho x_i - \delta_3 \frac{p_i x_i}{r_i}.
 \end{aligned} \tag{15}$$

The incomplete social interaction information includes in-degree and out-degree social structures generated by various IoT devices, where the in-degree indicates the number of other IoT devices affected by the IoT device. In contrast, the out-degree indicates the number of devices affecting that IoT device. The in-degree and out-degree distributions represent network interaction patterns with social network effects [37]. We assign in-degree $l \in \mathcal{L}$ and out-degree $O \in \mathcal{L}$ to each IoT device, where $\mathcal{L} = \{0, 1, \dots, L^{max}\}$ and L^{max} denotes the maximum probability value. Let $l : \mathcal{L} \rightarrow [0, 1]$ and $O : \mathcal{L} \rightarrow [0, 1]$ denote the marginal probability distributions for in-degree and out-degree, respectively. Consistency theory suggests that the average out-degree is equal to the average out-degree, i.e.,

$$\sum_{l=1}^L I(l)l = \sum_{O=1}^L H(O)O = \hat{O}, \tag{16}$$

where $\hat{O}$ denotes the average level of information about social interactions. Then, we have $E \left[\sum_{j \in N, j \neq i} x_j \right] = O_i \times \text{Avg}(x_{-i})$, where $\text{Avg}(x_{-i})$ denotes the average benefit level of the social neighborhood of user i . For an IoT device, the degree distribution of its chosen neighbor is $\bar{I}(l) = \frac{1}{\hat{O}} I(l)l$. The social neighbors of user i are randomly selected with an in-degree distribution $I(l)$ and an out-degree distribution $H(O)$. Denoting the resource request of user i with out-degree O and in-degree l as $x_i(O, l)$, it can be obtained that

$$\text{Ave}(f(x_{-i})) = \sum_l \bar{I}(l) \left(\sum_O H(O) x_i(O, l) \right). \tag{17}$$

Therefore, the expected utility of end user i with output degree O can be expressed as,

$$\begin{aligned}
 U'_i(\rho, x_i, x_{-i}) &= ax_i - bx_i^2 + \xi O \text{Ave}(x_{-i}) x_i + \delta_1 \kappa F_i^2 C_i x_i - \\
 &\quad \delta_2 \rho x_i - \delta_3 \frac{p_i x_i}{r_i}.
 \end{aligned} \tag{18}$$

B. Optimal Bandwidth Allocation Policy

In this subsection, we consider the bandwidth allocation strategy of terminal users under a given task offloading policy. By removing the constant term from the utility function of end users, the problem can be expressed as:

$$\begin{aligned}
 P3 : & \min_{b_i} \sum_{i \in N} \delta_3 \frac{p_i x_i}{r_i(b_i)}. \\
 s.t. \quad & C1 : 0 \leq b_i \leq 1, \\
 & C2 : \sum_{i \in N} b_i \leq 1,
 \end{aligned} \tag{19}$$

where $r_i(b_i) = b_i B \log(1 + \frac{p_i g_i}{\sigma^2})$.

To obtain the solution to Problem P3, we present the following theorem.

Theorem 1: Given the task offloading strategy for end users, the optimal bandwidth allocation strategy is

$$b_i^* = \begin{cases} 0, & x_i = 0 \\ \hat{b}, & 0 < \hat{b} < 1 \\ 1, & \hat{b} \geq 1 \end{cases}, i \in N. \quad (20)$$

where $\hat{b} = \frac{1}{\sum_{i \in N} \sqrt{\frac{p_i x_i}{\log(1 + \frac{p_i g_i}{\sigma^2})}}} \sqrt{\frac{p_i x_i}{\log(1 + \frac{p_i g_i}{\sigma^2})}}$.

Proof: First, it is known that problem P3 is a convex optimization problem, which can be solved using a partial Lagrange function. The corresponding Lagrange function can be obtained as

$$L(b_i, \theta) = \sum_{i \in N} \delta_3 \frac{p_i x_i}{b_i B \log(1 + \frac{p_i g_i}{\sigma^2})} + \theta \left(\sum_{i \in N} b_i - 1 \right). \quad (21)$$

where θ is a non-negative Lagrange multiplier.

Then, we use the KKT conditions to find the optimal bandwidth resource allocation scheme b_i^* . Specifically, we take the derivative of $L(f, \theta)$ with respect to B and set the result equal to 0, i.e., $\frac{\partial(L(b_i, \theta))}{\partial b_i} = 0$. The optimal computing resource allocation scheme can be expressed as

$$b_i = \sqrt{\frac{\delta_3 p_i x_i}{\theta B \log(1 + \frac{p_i g_i}{\sigma^2})}}. \quad (22)$$

It can be observed that θ must be greater than 0 in (22), so $\sum_{i \in N} b_i = 1$. Substituting b^* into constraint C2 of problem P3, one obtains

$$\sum_{i \in N} \sqrt{\frac{\delta_3 p_i x_i}{\theta B \log(1 + \frac{p_i g_i}{\sigma^2})}} = 1. \quad (23)$$

By solving (23), the optimal value of θ can be obtained, which can be specifically expressed as

$$\theta = \left(\sum_{i \in N} \sqrt{\frac{\delta_3 p_i x_i}{B \log(1 + \frac{p_i g_i}{\sigma^2})}} \right)^2 \quad (24)$$

Based on (22) and (24), the optimal bandwidth allocation strategy b_i can be obtained, which is formulated as

$$b_i = \frac{1}{\sum_{i \in N} \sqrt{\frac{p_i x_i}{\log(1 + \frac{p_i g_i}{\sigma^2})}}} \sqrt{\frac{p_i x_i}{\log(1 + \frac{p_i g_i}{\sigma^2})}}. \quad (25)$$

By considering the value range of the bandwidth allocation strategy, the optimal bandwidth allocation strategy b^* can be obtained. ■

Based on the analysis in Theorem 1, the optimal bandwidth allocation for the end user can be obtained.

C. Offloading Policy Based on Stackelberg Game

For the case of incomplete information about social interactions between IoT devices, we analyse the Bayesian Nash equilibrium for end users and give the following theorem.

Theorem 2: Considering a dynamic policy with a fixed number of users, for each user, the utility function satisfies equation (6). There exists a unique Nash equilibrium for the end users.

Proof: According to the benefit function in (18), the expected utility function of end user i can be expressed as,

$$U'_i(\rho, x_i, x_{-i}) = \xi O Ave(x_{-i}) x_i + a - b x_i^2 + \delta_1 \kappa F_i^2 C_i x_i - \delta_2 \rho x_i - \delta_3 \frac{p_i x_i}{r_i}. \quad (26)$$

Firstly, we can obtain the first derivative of U' concerning x as

$$\frac{\partial(U'_i(x_i, x_{-i}))}{\partial x_i} = \xi O Ave(x_{-i}) + a - 2b x_i + \delta_1 \kappa F_i^2 C_i - \delta_2 \rho - \frac{\delta_3 p_i}{r_i}. \quad (27)$$

Based on the above equation, we can obtain the second derivative of U' concerning x as,

$$\frac{\partial(U_i'^2(x_i, x_{-i}))}{\partial x_i^2} = -2b. \quad (28)$$

It can be observed that the second-order derivative of the end user's utility function U_i concerning x_i is negative. The utility function U_i is strictly concave. Therefore, there exists a unique Nash equilibrium point for end users. ■

Based on theorem 2, it can be seen that there is a unique Nash equilibrium for end users. The Nash equilibrium solution for end users is shown in the following theorem.

Theorem 3: In the game, the optimal task offloading strategy of the end user as a follower is denoted as

$$x_i^* = \max \left\{ [\Pi_i - \Upsilon_i \rho_i]^+, D_i \right\}, \quad (29)$$

where

$$\begin{aligned} [\Pi_i - \Upsilon_i \rho_i t]^+ &= \max(0, \Pi_i - \Upsilon_i \rho_i), \\ \Pi_i &= \frac{\Gamma}{2b} + \frac{\Lambda \Gamma}{2b - \xi \tilde{O}}, \\ \Upsilon_i &= \frac{\delta_2}{2b} + \frac{\Lambda \delta_2}{2b - \xi \tilde{O}}, \\ \Gamma_i &= a + \delta_1 \kappa F_i^2 C_i - \delta_3 \frac{p_i}{r_i}, \\ \Lambda_i &= \frac{\xi O}{2b}. \end{aligned} \quad (30)$$

Proof: To obtain a Bayesian equilibrium solution for the end user, setting $\frac{\partial(U'_i(\rho, x_i, x_{-i}))}{\partial x_i} = 0$ yields that

$$a - 2b x_i + \xi O Ave(x_{-i}) + \delta_1 \kappa F_i^2 C_i - \delta_2 \rho - \delta_3 \frac{p_i}{r_i} = 0. \quad (31)$$

By solving the above equation, we can obtain the optimal solution, which can be expressed as

$$x_i^* = \frac{a + \xi O Ave(x_{-i}) + \delta_1 \kappa F_i^2 C_i - \delta_2 \rho - \delta_3 \frac{p_i}{r_i}}{2b} + \quad (32)$$

According to (17), we can obtain

$$\begin{aligned}
 & E(f(O', l') | (O', l') \in L) \\
 &= \sum_{l'}^L \bar{I}(l') (\sum_{k'}^K H(O') x(O', l')) \\
 &= \frac{\sum_{l'}^L \bar{I}(l') \sum_{k'}^K H(O') (\xi O E(f(O'', l'') | (O'', l'') \in L) +}{2b} \\
 & \quad \frac{a + \delta_1 \kappa F_i^2 C_i - \delta_2 \rho - \delta_3 \frac{p_i}{r_i}}{2b} \\
 &= \frac{\xi \tilde{O} E(f(O'', l'') | (O'', l'') \in L + a + \delta_1 \kappa F_i^2 C_i - \delta_2 \rho - \delta_3 \frac{p_i}{r_i}}{2b},
 \end{aligned} \tag{33}$$

where $\tilde{O} = \sum_{l'}^L \bar{I}(l') \sum_{k'}^K H(O') O$.

Since $E(f(O', l') | (O', l') \in L) = E(f(O'', l'') | (O'', l'') \in L)$, we can obtain an expression for $E(f(O', l') | (O', l') \in L)$, which can be expressed as

$$E(f(O', l') | (O', l') \in L) = \frac{a + \delta_1 \kappa F_i^2 C_i - \delta_2 \rho - \delta_3 \frac{p_i}{r_i}}{2b - \xi \tilde{O}}. \tag{34}$$

Based on the above equation, we can obtain

$$\begin{aligned}
 x_i^* &= \frac{a + \delta_1 \kappa F_i^2 C_i - \delta_2 \rho - \delta_3 \frac{p_i}{r_i}}{2b} + \\
 & \quad \frac{\xi O}{2b} \frac{a + \delta_1 \kappa F_i^2 C_i - \delta_2 \rho - \delta_3 \frac{p_i}{r_i}}{2b - \xi \tilde{O}}.
 \end{aligned} \tag{35}$$

For simplicity, we define

$$\begin{aligned}
 \Gamma_i &= a + \delta_1 \kappa F_i^2 C_i - \delta_3 \frac{p_i}{r_i}, \\
 \Lambda_i &= \frac{\xi O}{2b}, \\
 \Pi_i &= \frac{\Gamma_i}{2b} + \frac{\Lambda_i \Gamma_i}{2b - \xi \tilde{O}}, \\
 \Upsilon_i &= \frac{\delta_2}{2b} + \frac{\Lambda_i \delta_2}{2b - \xi \tilde{O}}.
 \end{aligned} \tag{36}$$

Based on (36), x_i^* can be rewritten as

$$x_i^* = \Pi_i - \Upsilon_i \rho_i \tag{37}$$

Considering the constraints of the task offloading strategy, we can obtain the optimal offloading policy for the end user denoted as x_i^* . ■

D. Resource Pricing Policy Based on Stackelberg Game

In the case of incomplete information about social interactions between end users, we analyze the Bayesian Nash equilibrium in the leader game and give relevant theorems.

Theorem 4: Consider a fixed number of end users with a utility function satisfying (12) for the edge service provider, then an optimal pricing strategy for the edge service provider exists and is unique.

Proof: Firstly, knowing the task offloading size of the end user, the utility function of the edge service provider can be expressed as

$$U_e = \sum_{i \in N} (\rho_i - \varepsilon) (\Pi_i - \Upsilon_i \rho_i). \tag{38}$$

Algorithm 1: Bayesian Stackelberg-based Resource Pricing and Offloading Solution Algorithm.

```

Initialize number of iterations  $k = 0$ ;
Initialize precision threshold  $\zeta_1$  and  $\zeta_2$ ;
Initialize bandwidth allocation strategy  $b_i$ ;
Initialize resource price strategy  $\rho(k)$ ;
Set up  $\rho(k)$ , where  $0 \leq \rho(k) \leq \rho^{max}$ ;
Set up  $s(k)$ , where  $0 \leq s \leq \frac{S}{K}$ ;
while  $U^e(k+1) - U^e(k) \geq \zeta_1$  do
     $j = 0$ ;
    while  $b(j+1) - b(j) \geq \zeta_2$  do
        for  $i = 1$  to  $N$ 
            End users  $i$  calculate  $x_i$  based on social
            information, resource pricing, and bandwidth
            allocation strategy through (29);
            Obtain the optimal bandwidth allocation
            strategy through Theorem 1;
        End for
         $j = j + 1$ ;
    end while
     $\rho(k+1) = \rho(k) + \lambda \nabla U_e(\rho_i)$ ;
     $k = k + 1$ ;
end while
optimal  $x_i^*, i \in N$ ,  $\rho^*$ , and  $b_i^*, i \in N$  is obtained;

```

Finding the first-order derivative of U_e concerning ρ yields that

$$\frac{\partial U_e}{\partial \rho} = \Pi_i - 2\Upsilon_i \rho - \varepsilon \Upsilon_i. \tag{39}$$

The second-order derivative of U_e concerning ρ can be expressed as

$$\frac{\partial^2 U_e}{\partial \rho^2} = -2\Upsilon_i. \tag{40}$$

It can be observed that the second derivative of the utility function U_e of edge service providers with respect to ρ is negative.

Based on the above analysis, the utility function U_e of the Edge service provider is a strictly concave function, which proves the existence of a Nash equilibrium. ■

Based on Theorem 4, it can be seen that problem P2 is a convex optimization problem.

E. Dynamic Solution Algorithm

Based on the above game equilibrium analysis, a Bayesian Stackelberg-based Nash equilibrium solving algorithm is proposed, as shown in Algorithm 1. First, we set the initial parameters, which mainly include the initial pricing, bandwidth allocation strategy, and calculation precision. Second, it calculates the task offloading strategy based on the resource pricing, bandwidth allocation strategy, and social information provided by the edge service provider.

Secondly, end users formulate task offloading strategies based on bandwidth allocation strategies, social information, and resource pricing provided by edge service providers. Based on the new task offloading strategy, a new bandwidth

allocation strategy is obtained through Theorem 1. When the bandwidth resource allocation strategy meets the precision threshold, the end user stops updating the task offloading strategy, i.e., $b(j+1) - b(j) \leq \zeta_2$, where ζ_2 is the precision threshold.

Then, edge service providers update prices based on the task offloading strategy of end users. The resource pricing updates of edge service providers can be expressed as

$$\rho(k+1) = \rho(k) + \lambda \nabla U_e(\rho), \quad (41)$$

where λ denotes the update step size.

After completing the resource price update, the edge service provider broadcasts the updated prices to all end users, performing the $(t+1)$ iteration, i.e., end users update their task offloading strategies based on the new prices and recalculate their bandwidth allocation strategies. Accordingly, the final stopping condition is defined as

$$\|U^e(\rho(k+1), s(k+1)) - U^e(\rho(k), s(k))\| \leq \zeta_1, \quad (42)$$

where ζ_2 denotes the stop threshold.

F. Algorithm analysis

In this subsection, we analyze the complexity and convergence of Algorithm 1.

1) *Algorithm complexity*: In Algorithm 1, we define the first while loop as the outer loop, and the second while loop in Algorithm 1 as the inner loop. The computational complexity of the outer loop is $\log(1/\zeta_1)$, and the computational complexity of the inner loop is $\log(1/\zeta_2)$ [33]. Meanwhile, Algorithm 1 needs to consider the number of users when calculating the task strategy for end users. Therefore, the computational complexity of Algorithm 1 is $O(\log(1/\zeta_1) \log(1/\zeta_2) N)$.

2) *Algorithm convergence*: for the convergence of Algorithm 1, we give the following theorem.

Theorem 5: After a limited number of iterations, the results obtained by Algorithm 1 can converge to a stable value, and the strategies of edge service providers and end users reach a Nash equilibrium.

Proof: The convergence of Algorithm 1 depends on the strategy updates of the end users and the edge service provider. When both players in the game reach a Nash equilibrium state, the algorithm converges to a stable value. Theorems 2 and 4 guarantee the existence and uniqueness of the solution to the game defined in this paper. In addition, given the task offloading strategy of the end users, problem P3 is a convex optimization problem, i.e., there exists a unique solution. Based on Theorem 4, it is known that the resource pricing problem is a strictly concave maximization problem, and the gradient-based method in Algorithm 1 will converge to the optimal solution of this problem. Assuming that the accuracy threshold is sufficiently small, when the iterative difference $U^e(k+1) - U^e(k)$ of the edge service provider is within the threshold ζ_1 , the strategies of the end user and the edge service provider are stable, so the algorithm convergence to a stable value. ■

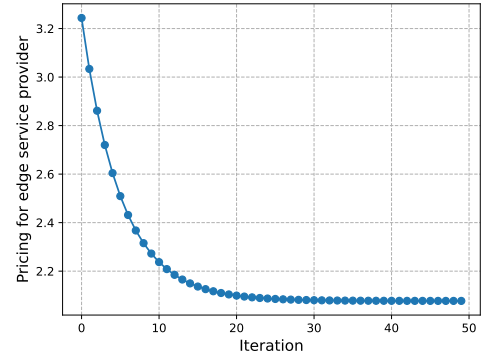


Fig. 2: Illustration for the user task offloading strategy of the proposed algorithm.

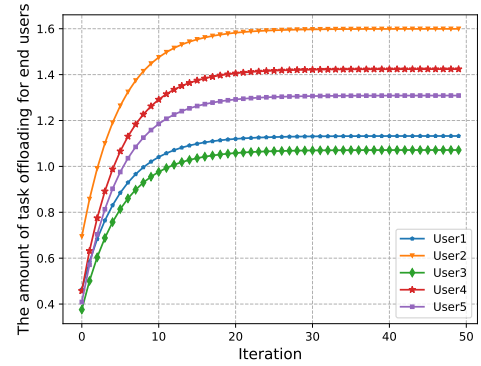


Fig. 3: Illustration for the pricing strategy of the edge service provider of the proposed algorithm.

V. NUMERICAL RESULTS AND PERFORMANCE ANALYSIS

In this section, we first present the simulation parameters and then evaluate the performance of computing services based on the Bayesian Stackelberg game in the IoT.

A. Parameter Setting

We consider a social network with N end users, whose in-degrees and out-degrees follow normal distributions $\mathcal{N}(\mu_1, \nu_1)$ and $\mathcal{N}(\mu_2, \nu_2)$, respectively, where $\mu_1 = \mu_2 = 7$ and $\nu_1 = \nu_2 = 1.2$. The end users are randomly distributed within the service range of the edge service provider, where the computing power of the end device is 0.1 GHz, transmission power is 0.2W [38], the data size of each computing task are randomly selected from $[2, 2.5]$ Mbits. The CPU cycles required for each unit task are 500 Megacycles. The bandwidth resources of edge service provider is 20 MHz [34]. The channel gain is denoted as $1/d^w$, where d denotes the distance from the end device to the edge device, and w denotes the path loss factor, which is set to 4 [43]. The parameters a and b of the end users are randomly selected from uniform distributions $[3.5, 4.5]$ and $[0.7, 1.2]$, respectively. The default parameters are set as: $\delta_1 = \delta_2 = \delta_3 = 1$, $\rho^{max} = 5$, $\sigma^2 = -100$ dbm [38], and $\kappa = 10^{-26}$. It is worth noting that some of the parameters can be varied depending on the evaluation scenario.

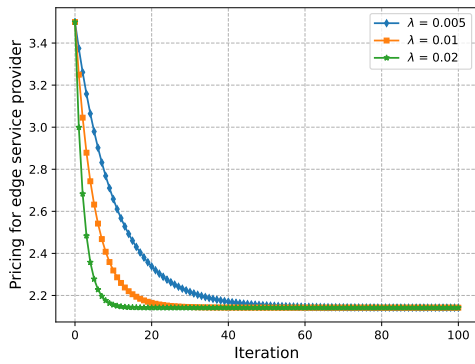


Fig. 4: Illustration for the convergence of the proposed algorithm under different update steps.

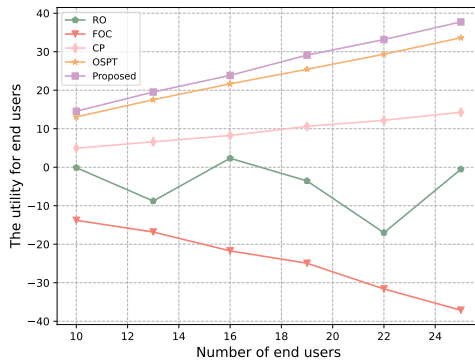


Fig. 5: Comparison of user utility under different numbers of end users.

1) *Full offloading computing (FOC)*: In this scheme, end users no longer consider the resource pricing strategy of edge service providers and choose to offload all tasks to edge service providers for processing.

2) *Random offloading (RO)*: In this strategy, the task offloading strategy for all end users is randomly generated, where the amount of data for task offloading is randomly selected from $[0, D_i]$, $i \in N$.

3) *Constant price (CP)*: In this strategy, edge service providers set resource pricing to a fixed value ρ , and the task offloading strategies of end users are still influenced by resource pricing and other users. [45].

4) *Offloading the Same Percentage of Tasks (OSPT)*: Similar to [46], in this strategy, it is assumed that the pricing strategy of the edge service provider is optimal, but all end users offload the same percentage of computation tasks.

B. Performance Evaluation

Fig. 2 illustrates the trend of resource pricing for edge service providers as the number of iterations increases. The resource pricing decreases with each iteration and eventually converges to a steady state. As algorithms are iterated, edge service providers will gradually lower their quotes to attract end users, thereby stimulating more task offloading activities. However, lower pricing may increase the service costs of edge service providers, thereby potentially reducing service utility. To address this situation, edge service providers will

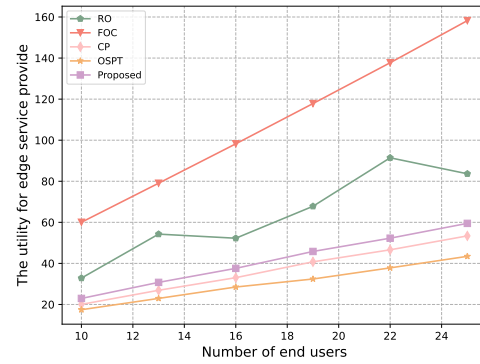


Fig. 6: Comparison of the utility of the edge service provider under different numbers of end users.

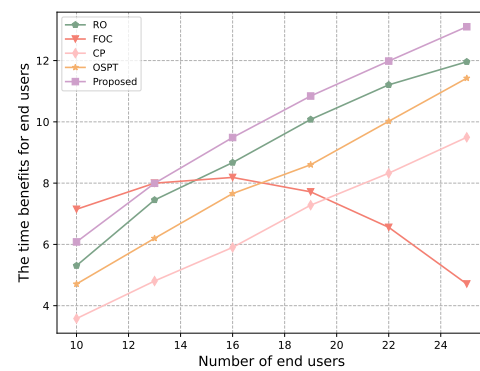


Fig. 7: Comparison of time benefits under different numbers of end users.

lower their quotes to strike a balance between profitability and service quality.

Fig. 3 illustrates the trend of end-user offloading service requests with an increasing number of iterations. As the iterations progress, the number of task offloading gradually increases and eventually converges to a steady state. When the task offloading size of end users and the resource pricing of edge service providers both reach a stable state, the game reaches a stable state. This indicates that both end users and edge service provider have achieved their ideal utility values, and proves the existence of a Stackelberg equilibrium in this case.

Fig. 4 illustrates the convergence rate of the proposed algorithm under various update step sizes. It can be observed that as the algorithm iterates, it eventually converges to a stable value under different update step sizes. This indicates that the algorithm has good convergence properties. In addition, it can be observed that all three curves exhibit a clear decreasing trend. Among them, the curve corresponding to $\lambda = 0.02$ shows the fastest initial decline and reaches a steady state earlier. $\lambda = 0.01$ follows; while $\lambda = 0.005$ demonstrates the slowest decrease and requires a longer convergence process. This is because a smaller update step size leads to slower progress in each iteration, thereby requiring more updates for the algorithm to reach a stable state.

Fig. 5 illustrates the trend of end-user utility under different strategies as the number of end users increases. It can be

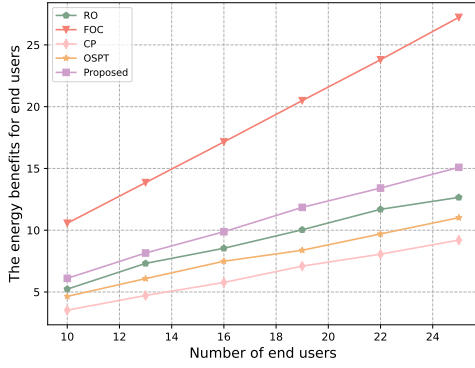


Fig. 8: Comparison of energy benefits under different numbers of end users.

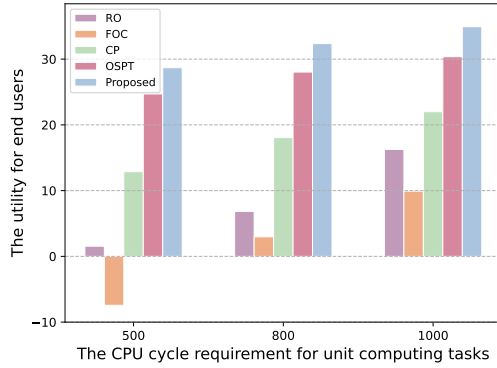


Fig. 9: Comparison of energy benefits under different numbers of end users.

observed that the OSPT strategy, the CP strategy, and the proposed strategy all show a gradual increase in user utility with the growth in the number of users. As the number of end users increases, more tasks are offloaded by users, which leads to a continuous decline in user utility under the FOC strategy. This is due to its inability to accommodate increasing offloading demands efficiently. The RO strategy exhibits considerable fluctuations because it relies on randomly generated policies. At the same time, it can be observed from the figure that the strategy proposed in this paper exhibits the optimal user utility under various numbers of users, and the utility value increases linearly with the number of users. This is because, as the number of end users increases, the strategy proposed in this paper can improve the overall utility by reasonably formulating the task offloading strategy for end users. This also demonstrates that the strategy proposed in this paper can protect the interests of end users.

Fig. 6 illustrates the variation in the utility of the edge service provider under different strategies as the number of end users increases. It can be observed that, as the number of end users grows, the utility of the edge service provider increases under all strategies. The FOC strategy exhibits the best performance, with its utility value far exceeding that of other strategies and increasing linearly with the growth in the number of users. This is because users offload all tasks to the edge service provider, thereby increasing the revenue of the edge service provider. In contrast, the strategy proposed in this

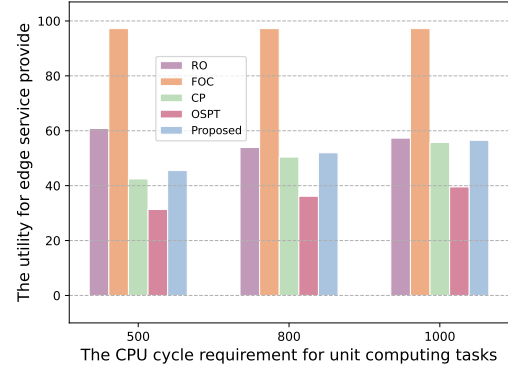


Fig. 10: Comparison of energy benefits under different numbers of end users.

paper has obvious advantages over the CP strategy and OSPT strategy while maintaining stable growth. This indicates that the proposed strategy not only enables end users to achieve higher utility but also allows the edge service provider to obtain improved utility.

Fig. 7 illustrates the trend of time benefits for end users as the number of end users increases under different strategies, where the time benefit for end users is calculated as $T_i^b = t_i^{floc} - \max\{t_i^e, t_i^{loc}\}$, T_i^b , T_i^b denotes the time benefit of end user i , t_i^{floc} denotes the execution delay when all tasks of end user i are processed locally. As shown in the figure, the strategy proposed in this paper exhibits the optimal time benefit across all user scales. It exhibits a sustained upward trend as the number of users increases. This indicates that the strategy proposed in this paper has a good task scheduling strategy and can effectively reduce the task processing delay for end users. It is worth noting that as the number of users increases, the time benefits of the FOC strategy decrease rather than increase. This is because, as user tasks increase, the edge service provider cannot allocate appropriate computing resources to each user, thereby reducing the time benefits for end users.

Fig. 8 illustrates the variation in end-user energy benefits under different strategies as the number of end users increases, where the energy benefit is calculated as $\kappa F_i^2 C_i x_i - p_i x_i / r_i$. It can be observed that the energy benefits of all strategies increase with the number of users. This is because users offload their tasks to the edge service provider, thereby reducing their local computation energy consumption. Meanwhile, it can be observed that the proposed strategy achieves better energy benefits compared to the other strategies. This also indicates that the strategy proposed in this paper can balance user energy consumption and latency, not only reducing the computing latency of end users, but also achieving good energy consumption benefits.

Fig. 9 illustrates the utility of end users under varying CPU cycle requirements for unit computing tasks. A comparison of different methods reveals that the strategy proposed in this paper consistently achieves the highest end-user utility across all CPU cycle requirements, with the magnitude of improvement becoming more pronounced as the CPU cycles increase. This indicates that the proposed strategy is capable of main-

taining superior performance even under high computational resource demands. This indicates that the strategy can maintain superior performance even under high computational resource requirements. As the CPU cycle requirements increase, the utility values of all strategies exhibit an overall upward trend, albeit with differing growth rates. The strategy proposed in this paper effectively enhances end-user utility across tasks of varying computational complexities. This demonstrates its robustness and superiority in resource allocation and task scheduling, offering a more effective solution for addressing diverse computational workloads.

Fig. 10 presents the utility of the edge service provider under varying CPU cycle requirements for unit computing tasks. It can be observed that the FOC strategy achieves the highest utility. Although the FOC strategy holds an advantage in terms of edge service provider benefits, the strategy proposed in this paper attains medium-to-high revenue while simultaneously delivering significant improvements in user utility. This demonstrates the advantages of the proposed strategy in balancing the interests of multiple parties and optimizing resource allocation.

VI. CONCLUSION

In this paper, we study an MEC network consisting of an edge service provider and multiple IoT end users. To achieve reasonable resource pricing, resource allocation, and task offloading decisions, we first formulate the computation offloading problem between the edge service provider and end users as a Stackelberg game. In the task offloading process, end users consider not only the resource pricing of edge service providers, but also social benefits arising from social interactions with other user devices. In addition, bandwidth allocation strategies are optimized to reduce transmission energy consumption for end users. Second, to maximize the utility of the edge service provider and end users, backward induction is used to analyze the proposed game. Then, we analyze the existence and uniqueness of the Stackelberg equilibrium and design an iterative algorithm to solve the strategies of both parties to achieve the equilibrium state of the game. Finally, simulation results verify that our proposed scheme can provide an appropriate task offloading solution for IoTs.

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